Indian Institute of Technology, Hyderabad



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Course Name

Mechanics of Solids

DIC Studio

A Virtual Digital Image Correlation Lab

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Project Documentation

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Project Links

Recommended: Download the DIC Studio Application

We recommend downloading and using the standalone DIC Studio application. It runs locally on your computer and does not require my laptop to be turned on. Separate application packages for macOS and Windows are included with the project submission.

GitHub Repository:

DIC-Mechanics-of-Solids

Project Website (works only when my laptop is turned on):

DIC Studio - Virtual Solid Mechanics Laboratory

Contents

1 Project Overview **1**
- 1.1 What the project does . . . 1
- 1.2 Implemented experiments . . . 1
- 1.3 End-to-end workflow . . . 1

2 Problem Statement and Motivation **2**
- 2.1 The gap between a mechanics formula and a measurement . . . 2
- 2.2 Context in digital image correlation . . . 2
- 2.3 The problem being solved . . . 2
- 2.4 Why use a virtual experiment? . . . 2

3 Analytical Solid Mechanics Models **3**
- 3.1 Coordinates, material law and assumptions . . . 3
- 3.2 Axial bar under uniform tension . . . 3
- 3.3 End-loaded cantilever benchmark . . . 3
- 3.4 Pure bending by moments at both ends . . . 4
- 3.5 Infinite plate containing a circular hole . . . 5
- 3.6 Square-section torsion . . . 5

4 Synthetic Speckle Images and Camera Modelling **8**
- 4.1 Reference texture and physical calibration . . . 8
- 4.2 Why inverse warping is used . . . 8
- 4.3 Initial guess and fixed-point iteration . . . 8
- 4.4 Cubic B-spline interpolation . . . 9
- 4.5 Camera degradation and quantization . . . 9
- 4.6 Material and interpolation support . . . 10

5 Post-Capture Image Denoising **11**
5.1 Purpose and position in the pipeline . . . 11
5.2 Gaussian filtering . . . 11
5.3 Local Wiener filtering . . . 11
5.4 How filter quality is evaluated . . . 12

6 Digital Image Correlation for Flat Specimens **13**
6.1 Subset-based measurement . . . 13
6.2 Integer-pixel initialization by ZNCC . . . 13
6.3 Six-parameter affine shape function . . . 13
6.4 Normalized nonlinear least-squares refinement . . . 14
6.5 Acceptance and support after refinement . . . 14
6.6 Independent forward/backward verification . . . 14
6.7 Complete matching procedure . . . 15

7 Recovering Strain and Inferring Stress **16**
7.1 Why differentiating displacement is difficult . . . 16
7.2 Weighted quadratic displacement fitting . . . 16
7.3 Converting image derivatives into physical strain . . . 16
7.4 Choosing the fitting radius . . . 17
7.5 Plane-stress constitutive conversion . . . 17
7.6 Principal and equivalent stress . . . 17

8 Stereo DIC for Square-Beam Torsion **18**
8.1 Why two-dimensional DIC is insufficient . . . 18
8.2 Calibrated pinhole projection and image generation . . . 18
8.3 Independent image correspondences . . . 18
8.4 Triangulation of reference and deformed surfaces . . . 19
8.5 Geometric and consistency checks . . . 19
8.6 Surface strain with rigid-rotation invariance . . . 19
8.7 Analytical validation and stress limits . . . 20

9 Implementation and Interactive Laboratory **21**
9.1 Separation of responsibilities . . . 21
9.2 User workflow and numerical limits . . . 21
9.3 Views and interpretation . . . 21
9.4 Reproducible outputs . . . 22

10 Validation, Error Metrics and Robustness Protocol **23**
10.1 Ground truth at matching locations . . . 23
10.2 Vector and combined-component errors . . . 23
10.3 Coverage and the accepted population . . . 23
10.4 Hole-edge concentration versus sampled interior maximum . . . 24
10.5 Paired imaging and denoising comparisons . . . 24
10.6 One-factor-at-a-time robustness study . . . 24
10.7 Independent software and mechanics checks . . . 24

11 Numerical Results and Discussion **25**
11.1 Baseline experiment protocol . . . 25
11.2 Flat displacement, strain and stress recovery . . . 25
11.3 The hole boundary and concentration interpretation . . . 26
11.4 Denoising: an experiment-dependent tradeoff . . . 26
11.5 Imaging robustness across five factors . . . 27
11.6 Square torsion and visible-face stereo recovery . . . 27
11.7 What the fresh evidence establishes . . . 29

12 Limitations, Future Work and Conclusions **30**
12.1 Model and measurement limitations . . . 30
12.2 Priorities for future development . . . 30
12.3 Conclusions . . . 30

References **31**

A Notation and Units **32**

1 Project Overview

1.1 What the project does

DIC Studio is an interactive virtual laboratory that connects solid mechanics with optical deformation measurement. A user selects a specimen, material properties, dimensions and load. The program computes an analytical deformation, renders synthetic camera images of a speckled surface, and measures motion from those images using digital image correlation (DIC). Measured displacements are then differentiated to obtain strain. For flat specimens, known elastic properties convert that strain into stress. Finally, measured and analytical fields are compared at the same reference locations.

The imposed deformation is known to the image generator but withheld from DIC. The laboratory therefore shows both the mechanical response and the accuracy of its image-based recovery. Stress inference is a modelling step after optical displacement measurement [2].

1.2 Implemented experiments

Specimen	Mechanical benchmark	Image-based outputs
Axial bar	Uniform uniaxial plane-stress tension	In-plane displacement, strain and inferred stress
Cantilever	Saint-Venant end-shear elasticity field	In-plane displacement, strain and inferred stress
Plate with a hole	Infinite-plate Kirsch solution in a finite observation window	Interior displacement, strain and inferred stress
Pure bending	Equal and opposite end moments	In-plane displacement, strain and inferred stress
Square beam in torsion	Free-warping Saint-Venant torsion	Visible-face 3D displacement and surface strain from stereo images

Table 1: The five selectable experiments and what is actually recovered from images.

It is an analytical-benchmark and DIC laboratory, not a general FEM solver. Its 3D preview visualizes the prescribed analytical field.

1.3 End-to-end workflow

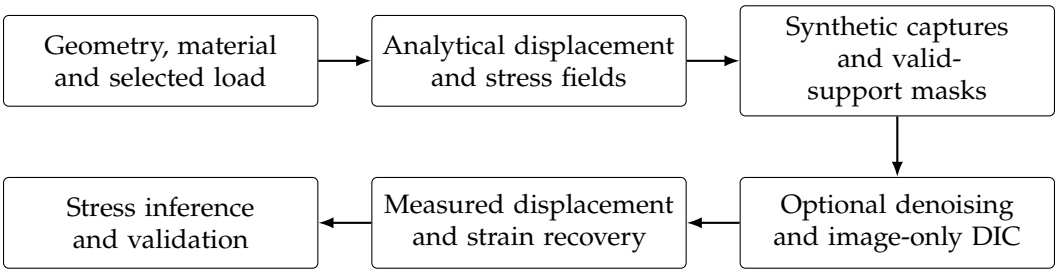


Figure 1: The flat-specimen pipeline. Analytical truth is used to generate captures and later evaluate errors; it does not provide DIC motion guesses. Stereo additionally uses known camera calibration.

2 Problem Statement and Motivation

2.1 The gap between a mechanics formula and a measurement

Classical solid mechanics provides displacement, strain and stress fields under idealized loads and boundary conditions. A camera, however, records a grid of intensities. It does not directly record displacement derivatives or stress. The practical problem is therefore to establish a transparent chain from images to mechanical quantities, quantify the errors introduced at each stage, and distinguish measurement failure from a physically small response.

Speckled images provide spatial information over a surface, but recovering motion introduces interpolation, optimization and differentiation errors. Stress inference magnifies strain error through elastic stiffness. A convincing deformation animation is therefore insufficient evidence of accurate recovered stress.

2.2 Context in digital image correlation

Digital image correlation (DIC) estimates surface motion by comparing image patterns between reference and deformed captures. Subset-based methods use the intensity pattern around each point to estimate its correspondence; interpolation and optimization allow the estimate to vary below one pixel. Strain is then obtained from spatial derivatives of the measured displacement, so errors in displacement can be amplified during differentiation.

This project uses local subsets with an affine displacement approximation, then estimates strain from a separate local fit to accepted subset-centre displacements. The resulting stress is inferred using the specified elastic material properties and plane-stress assumption; it is not measured directly from image intensity.

2.3 The problem being solved

For a specimen with prescribed geometry, elastic properties and loading, the project addresses the following question:

> How can deformation be independently recovered from synthetic camera images, converted into useful mechanical fields, and compared with known ground truth while exposing the effects of noise, spatial resolution and rejected measurements?

2.4 Why use a virtual experiment?

A virtual experiment reuses the same texture and load while changing only noise or filtering. It evaluates analytical values at the measured centres, retains failed matches and exposes unavailable strain fits.

This allows controlled study of noisy stress derivatives, fitting-radius effects, the unmeasured boundary gap around a hole, and why out-of-plane torsion needs stereo imaging.

3 Analytical Solid Mechanics Models

3.1 Coordinates, material law and assumptions

The flat specimens use Cartesian coordinates x to the right and y upward. Displacement is (u, v) in metres; stresses are in pascals. The material is homogeneous, isotropic and linearly elastic, with Young's modulus E, Poisson's ratio ν, and shear modulus

$$G = \frac{E}{2(1+\nu)}. \tag{1}$$

For small displacement gradients, the strain-displacement relations are

$$\varepsilon_{xx} = u_{,x}, \qquad \varepsilon_{yy} = v_{,y}, \qquad \gamma_{xy} = u_{,y} + v_{,x} = 2\varepsilon_{xy}. \tag{2}$$

Plane stress assumes $\sigma_{zz} = \tau_{xz} = \tau_{yz} = 0$. It does not set the transverse normal strain to zero. The compliance relations are

$$\varepsilon_{xx} = \frac{\sigma_{xx} - \nu\sigma_{yy}}{E}, \quad \varepsilon_{yy} = \frac{\sigma_{yy} - \nu\sigma_{xx}}{E}, \quad \gamma_{xy} = \frac{\tau_{xy}}{G}. \tag{3}$$

The implemented displacement and stress fields must satisfy the same constitutive relations. This is especially important for bending: mixing an approximate transverse beam displacement with an unrelated shear-stress formula would generate an inconsistent ground-truth pair.

3.2 Axial bar under uniform tension

Equilibrium and axial extension

For a uniform one-dimensional bar of area A and end force P, axial equilibrium gives $N = P$. Thus $\sigma = P/A$ and $u_{,x} = P/(AE)$. Setting $u(0) = 0$ yields

$$u(x) = \frac{Px}{AE}. \tag{4}$$

The two-dimensional image experiment includes the corresponding Poisson contraction. Writing $S = P/A$ for nominal stress,

$$u(x,y) = \frac{S}{E}x, \qquad v(x,y) = -\frac{\nu S}{E}y, \qquad (\sigma_{xx}, \sigma_{yy}, \tau_{xy}) = (S, 0, 0). \tag{5}$$

Substitution into Equation (2) gives $(S/E, -\nu S/E, 0)$, which agrees with Equation (3). Stress is uniform, so a uniform stress colour over the material surface is expected. The field fixes an axial reference at $x = 0$ while allowing lateral contraction; it does not represent a fully rigid two-dimensional clamp.

3.3 End-loaded cantilever benchmark

Consider a beam of length L, height h and thickness b, with $0 \le x \le L$, $-h/2 \le y \le h/2$ and $I = bh^3/12$. Positive P acts upward at the end. The implementation uses the following Saint-Venant plane-stress polynomial field:

$$\sigma_{xx} = -\frac{P(L-x)y}{I}, \qquad \sigma_{yy} = 0, \qquad \tau_{xy} = \frac{P}{2I}\left(\frac{h^2}{4} - y^2\right). \tag{6}$$

Its compatible displacement, with $q = P/(EI)$, is

$$u = -q\left[\left(Lx - \frac{x^2}{2}\right)y + \frac{(2+\nu)y^3}{6}\right], \tag{7}$$

$$v = q\left[\frac{x^2(3L-x)}{6} + \frac{(1+\nu)h^2 x}{4} + \frac{\nu(L-x)y^2}{2}\right]. \tag{8}$$

Why this field is mechanically compatible

Differentiation gives

$$u_{,x} = -q(L-x)y, \qquad v_{,y} = q\nu(L-x)y, \tag{9}$$

$$u_{,y} + v_{,x} = q(1+\nu)\left(\frac{h^2}{4} - y^2\right). \tag{10}$$

These agree with Equation (3) applied to Equation (6). In addition,

$$\sigma_{xx,x} + \tau_{xy,y} = \frac{Py}{I} - \frac{Py}{I} = 0, \qquad \tau_{xy,x} + \sigma_{yy,y} = 0, \tag{11}$$

so in-plane equilibrium is satisfied without body force. The top and bottom surfaces have $\sigma_{yy} = \tau_{xy} = 0$. The section resultants are

$$b\int_{-h/2}^{h/2} \tau_{xy}\,\mathrm{d}y = P, \qquad -b\int_{-h/2}^{h/2} y\sigma_{xx}\,\mathrm{d}y = P(L-x). \tag{12}$$

The loading is therefore a parabolic end shear traction with the correct force and bending moment. The displacement convention fixes the centre at the left reference section, but does not force every point of that section to remain immobile. Local stresses and displacements introduced by an actual rigid clamp are outside this benchmark.

3.4 Pure bending by moments at both ends

Let the signed constant bending moment be M, and define $\kappa_b = M/(EI)$ and $\xi = x - L/2$. A compatible field is

$$u = -\kappa_b \xi y, \qquad v = \frac{\kappa_b}{2}\left[\xi^2 - (L/2)^2 + \nu y^2\right], \tag{13}$$

$$\sigma_{xx} = -\frac{My}{I}, \qquad \sigma_{yy} = \tau_{xy} = 0. \tag{14}$$

Indeed $u_{,x} = -\kappa_b y$, $v_{,y} = \nu\kappa_b y$, and $u_{,y} + v_{,x} = 0$. The νy^2 term is required for constitutive compatibility. Positive M compresses the upper half and extends the lower half; $y = 0$ is the neutral axis.

The external end resultants are equal and opposite moments, with no transverse shear or net axial force. Setting $v(0,0) = v(L,0) = 0$ and $u(L/2,0) = 0$ removes arbitrary rigid motion; these are reference choices rather than two rigid clamps. Reversing M reverses the signed displacement and axial stress. Local moment-introduction effects, buckling and yielding are not modelled.

3.5 Infinite plate containing a circular hole

Stress solution and coordinate convention

A circular hole of radius a lies at the origin in an otherwise infinite plate under remote tension S along x. Let $r = (x^2+y^2)^{1/2}$, $\theta = \operatorname{atan2}(y,x)$, and $z = a^2/r^2$. The Kirsch stresses are [3]

$$\sigma_{rr} = \frac{S}{2}\left[1 - z + (1 - 4z + 3z^2)\cos 2\theta\right], \tag{15}$$

$$\sigma_{\theta\theta} = \frac{S}{2}\left[1 + z - (1 + 3z^2)\cos 2\theta\right], \tag{16}$$

$$\tau_{r\theta} = -\frac{S}{2}\left[1 + 2z - 3z^2\right]\sin 2\theta. \tag{17}$$

Writing $c = \cos\theta$ and $s = \sin\theta$, Cartesian stresses follow from rotation:

$$\sigma_{xx} = \sigma_{rr}c^2 + \sigma_{\theta\theta}s^2 - 2\tau_{r\theta}sc, \tag{18}$$

$$\sigma_{yy} = \sigma_{rr}s^2 + \sigma_{\theta\theta}c^2 + 2\tau_{r\theta}sc, \tag{19}$$

$$\tau_{xy} = (\sigma_{rr} - \sigma_{\theta\theta})sc + \tau_{r\theta}(c^2 - s^2). \tag{20}$$

The cited benchmark uses its own angle/loading convention; the expressions above state the convention actually used here.

Compatible polar displacement

With $k = (3-\nu)/(1+\nu)$ for plane stress, the implemented displacements are

$$u_r = \frac{S}{4G}\left\{ r\left[\frac{k-1}{2} + \cos 2\theta\right] + \frac{a^2}{r}\left[1 + (k+1)\cos 2\theta\right] - \frac{a^4}{r^3}\cos 2\theta \right\}, \tag{21}$$

$$u_\theta = \frac{S}{4G}\left\{ -r\sin 2\theta + (1-k)\frac{a^2}{r}\sin 2\theta - \frac{a^4}{r^3}\sin 2\theta \right\}, \tag{22}$$

and $u = u_r\cos\theta - u_\theta\sin\theta$, $v = u_r\sin\theta + u_\theta\cos\theta$. The hole interior is excluded from material and measurement masks. A bounded numerical extension inside it assists inverse-coordinate iteration only; it is not valid material ground truth.

Stress concentration and the observation window

At $r = a$, $\sigma_{rr} = \tau_{r\theta} = 0$ and

$$\sigma_{\theta\theta} = S(1 - 2\cos 2\theta), \qquad K_t = \frac{\max \sigma_{\theta\theta}}{S} = 3. \tag{23}$$

This peak occurs at the top and bottom of the hole. The displayed rectangle is a camera crop of the infinite-domain solution, not a finite rectangular plate solved with applied edge tractions. When a user specifies force, the interface converts it to equivalent remote stress using gross area hb. This does not change the infinite-domain assumption.

3.6 Square-section torsion

Why the circular formula cannot simply be reused

For a solid circular shaft, $J = \pi R^4/2$ and the shear stress magnitude is $|T|r/J$. A square section warps and has a different torsion constant; replacing r with a distance to a square boundary would

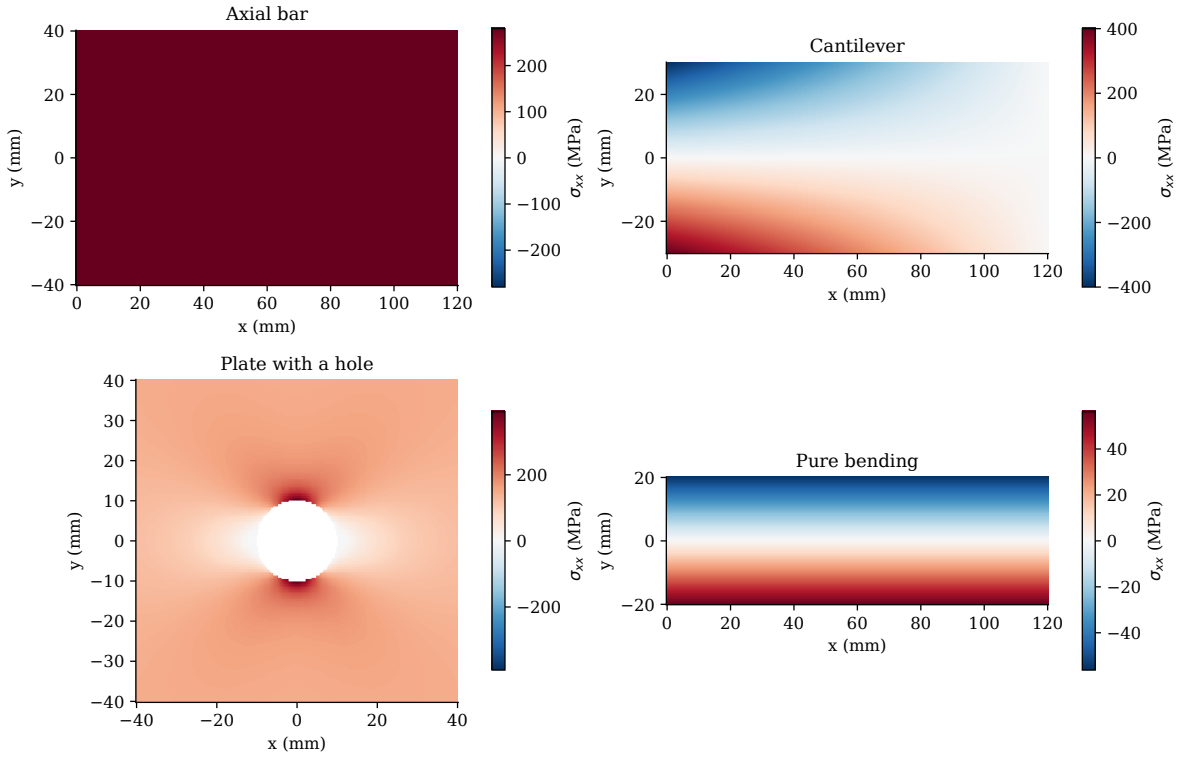


Figure 2: Analytical signed axial stress patterns for the four flat benchmarks. Each panel uses its own indicated MPa colour scale. These are model predictions, not recovered DIC fields.

not satisfy the lateral traction conditions. Saint-Venant torsion instead uses a cross-sectional stress function and free axial warping [4].

Let the square have side $s = 2a$ with $-a \le y, z \le a$. The torsion constant implemented with 80 odd series terms is

$$J = \frac{s^4}{3}\left[1 - \frac{192}{\pi^5}\sum_{n=1,3,\ldots,159}\frac{\tanh(n\pi/2)}{n^5}\right], \qquad \alpha = \frac{T}{GJ}. \tag{24}$$

Here α is the twist per unit length. A normalized Prandtl stress function ϕ satisfies

$$\nabla^2_{yz}\phi = -2, \qquad \phi = 0 \text{ on the square boundary}, \qquad \tau_{xy} = G\alpha\phi_{,z}, \quad \tau_{xz} = -G\alpha\phi_{,y}. \tag{25}$$

The associated warping function used in the code is

$$w(y,z) = -yz + \frac{32a^2}{\pi^3}\sum_{j=0}^{79}\frac{(-1)^j}{(2j+1)^3}\sin\left(\frac{(2j+1)\pi z}{2a}\right)\frac{\sinh[(2j+1)\pi y/(2a)]}{\cosh[(2j+1)\pi/2]}. \tag{26}$$

The stresses can also be expressed as $\tau_{xy} = G\alpha(w_{,y} - z)$ and $\tau_{xz} = G\alpha(w_{,z} + y)$. Exponentially scaled hyperbolic ratios avoid overflow in the high-order terms. Exact boundary coordinates enforce the known zero lateral traction of the retained stress derivatives.

Middle-section rotation reference

The reference angle is selected at the middle, so that

$$\theta(x) = \alpha(x - L/2), \qquad \theta(0) = -\alpha L/2, \quad \theta(L) = +\alpha L/2. \tag{27}$$

Both ends therefore rotate oppositely while the total relative twist is αL. This removes a rigid rotation from the visualization; it does not introduce a physical support at the middle.

For camera geometry the section is rotated using finite sine and cosine:

$$\boldsymbol{x}(x,y,z) = \begin{bmatrix} x + \alpha w(y,z) \\ y\cos\theta - z\sin\theta \\ y\sin\theta + z\cos\theta \end{bmatrix}. \tag{28}$$

The analytical stress remains the small-strain Saint-Venant stress. This finite-rotation geometric rendering is not a nonlinear constitutive solution. The equivalent stress for this shear-only stress state is $\sigma_{\mathrm{VM}} = \sqrt{3}\sqrt{\tau_{xy}^2 + \tau_{xz}^2}$. These bulk stresses remain analytical in the stereo workflow.

4 Synthetic Speckle Images and Camera Modelling

4.1 Reference texture and physical calibration

The specimen surface is represented by a reproducible grayscale speckle pattern. A seeded random generator places dark discs over a background intensity of 0.82. Antialiased edges produce fractional intensities rather than binary stair steps. Disc radii vary between $0.7r_s$ and $1.3r_s$, where r_s is the selected nominal speckle radius in pixels. The density input controls approximate expected coverage; disc overlap means it is not an exact black-pixel percentage.

Let an image have W columns and H rows, with physical bounds $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$. Metres per pixel are

$$s_x = \frac{x_{\max} - x_{\min}}{W - 1}, \qquad s_y = \frac{y_{\max} - y_{\min}}{H - 1}. \tag{29}$$

Column c and row r map to $x = x_{\min} + cs_x$, $y = y_{\max} - rs_y$. Therefore a physical displacement maps to image displacement as

$$d_c = u/s_x, \qquad d_r = -v/s_y. \tag{30}$$

The minus sign is essential because image rows increase downward. Horizontal and vertical pixel sizes need not be equal. Both images use the same camera pixel grid; material points move relative to that grid, rather than stretching the grid itself.

4.2 Why inverse warping is used

Let $\boldsymbol{X}$ denote a reference material coordinate and let $\boldsymbol{x}$ be its deformed location:

$$\boldsymbol{x} = \boldsymbol{X} + \boldsymbol{u}(\boldsymbol{X}). \tag{31}$$

A forward renderer would send each reference pixel to its deformed position. Since positions are generally fractional, some output pixels could receive multiple contributions and others none. Instead, inverse warping visits each output pixel and asks which reference point moved there. Conservation of the assumed surface texture gives

$$I_1^{\text{ideal}}(\boldsymbol{x}) = I_0\big(\boldsymbol{X}(\boldsymbol{x})\big). \tag{32}$$

This is a geometric sampling rule before camera blur, noise or brightness changes. The intensity at a deformed pixel is obtained from the original texture at the corresponding reference coordinate.

4.3 Initial guess and fixed-point iteration

Equation (31) is implicit in $\boldsymbol{X}$ when displacement varies spatially. The implementation starts from the zero-motion guess $\boldsymbol{X}^{(0)} = \boldsymbol{x}$ and updates

$$\boldsymbol{X}^{(k+1)} = \boldsymbol{x} - \boldsymbol{u}\big(\boldsymbol{X}^{(k)}\big). \tag{33}$$

Each step evaluates displacement at the current estimate of the original point and subtracts it from the fixed output location. It does not move the output pixel or fit image intensities. For an affine field $\boldsymbol{u}(\boldsymbol{X}) = A\boldsymbol{X} + \boldsymbol{b}$, the exact inverse is $(I + A)^{-1}(\boldsymbol{x} - \boldsymbol{b})$; iteration errors satisfy $\boldsymbol{e}^{(k+1)} = -A\boldsymbol{e}^{(k)}$. Small gradients make this a contraction. General nonuniform motion is not exactly inverted by a single $\boldsymbol{x} - \boldsymbol{u}(\boldsymbol{x})$ evaluation.

The code stops when the largest change, expressed in either pixel direction, is below 10^{-8} pixels. It allows at most 100 iterations and raises an error on nonconvergence. For example, in one dimension with $u(X) = 0.1X$ and output coordinate $x = 11$, the guesses are 11, 9.9, 10.01, 9.999, and so on, converging to $X = 10$.

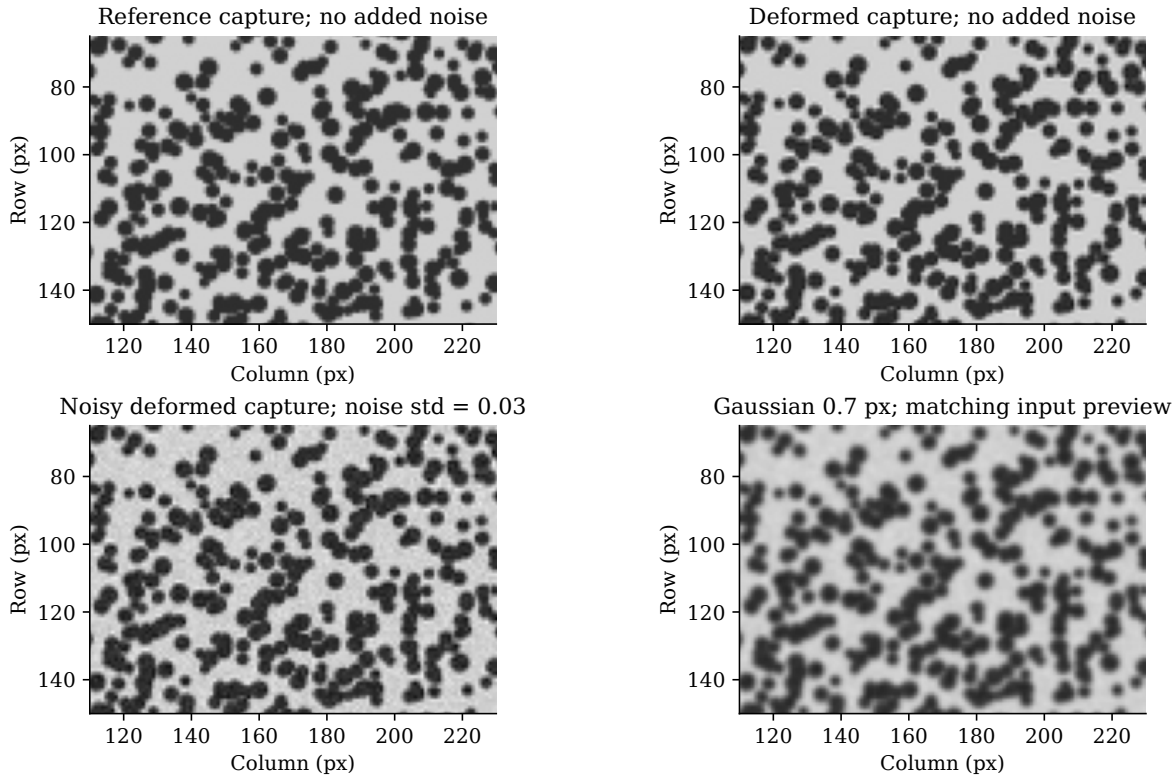


Figure 3: Representative images from the fresh cantilever experiments: raw reference and deformed captures, a noisy deformed capture, and a post-capture Gaussian-filtered input. Crops use identical image coordinates; intensity changes are not displacement vectors.

4.4 Cubic B-spline interpolation

The reference image stores grayscale intensities only at integer pixel coordinates. However, deformation maps pixels to subpixel locations, which generally do not coincide with the original pixel grid. Since intensity values at these locations are unavailable, interpolation is used to estimate them. This is essential for generating smooth, accurate deformed images and avoiding the gaps and aliasing that can occur when pixels are mapped directly.

The image intensity at an inverse-mapped coordinate (c^*, r^*) is evaluated using cubic B-spline interpolation through `map_coordinates(order=3)` [5]:

$$I_0(c^*, r^*) = \sum_i \sum_j a_{ij} \beta_3(c^* - i) \beta_3(r^* - j). \tag{34}$$

Here, β_3 is the cubic B-spline basis and a_{ij} are the spline coefficients. Its compact support means that each interior evaluation depends on four coefficients in each direction, or 16 coefficients in total. SciPy's default prefilter computes these coefficients from the original image; they are not simply the pixel intensities themselves. The DIC stage independently constructs a bicubic spline from the captured image for subpixel matching.

4.5 Camera degradation and quantization

For each ideal image, optional Gaussian blur first represents isotropic optical smoothing. A gain g and offset b_0 may then modify the deformed image. Independent noise fields n_0, n_1 are drawn

from $\mathcal{N}(0,\sigma_n^2)$. The captured normalized images are

$$I_0^{\mathrm{cam}} = Q_8\left[\mathcal{G}_{\sigma_b} * I_0^{\mathrm{ideal}} + n_0\right], \tag{35}$$

$$I_1^{\mathrm{cam}} = Q_8\left[g(\mathcal{G}_{\sigma_b} * I_1^{\mathrm{ideal}}) + b_0 + n_1\right], \tag{36}$$

with $Q_8(z) = \mathrm{round}(255\,\mathrm{clip}(z,0,1))/255$. Gaussian noise perturbs intensities and hence the correlation residuals. Gaussian blur removes high spatial frequencies and weakens fine texture, although mild smoothing can sometimes reduce sampling artefacts. It is not directional motion blur. Gain and offset are distinct from random noise, and clipping can destroy the ideal brightness-normalization invariance.

Noise standard deviation is measured on the normalized intensity scale: $\sigma_n = 0.01$ corresponds to 2.55 grayscale levels before clipping. Blur width is measured in pixels. Even with both inputs set to zero, interpolation and 8-bit rounding remain possible error sources.

4.6 Material and interpolation support

The reference mask excludes the hole; the deformed mask follows the inverse material coordinate and checks image support. Camera-blur footprints, denoising footprints and an additional interpolation margin are excluded from matching. Thus a pixel near a material boundary may be visually present but unavailable for a complete DIC subset. Synthetic geometry supplies these masks. A real camera experiment would need independent segmentation and calibration.

5 Post-Capture Image Denoising

5.1 Purpose and position in the pipeline

Denoising is an optional operation on the captured images immediately before matching. It attempts to suppress intensity noise while retaining enough spatial texture to localize a subset. It cannot restore detail removed by camera blur, recover saturated pixels reliably, or guarantee that a correlation peak corresponds to the correct material point. No clean image or analytical displacement is available to the preprocessing module.

The original 8-bit captures are preserved. Matching uses floating-point filtered arrays saved in `dic_input_images.npz`; the smoothed PNGs are 16-bit previews. They are not saved and reloaded as DIC inputs. Both images in a flat experiment, or all four images in stereo, receive the same selected filter type.

5.2 Gaussian filtering

The normalized two-dimensional Gaussian kernel is

$$K_{\sigma_d}(c,r) = \frac{1}{2\pi\sigma_d^2}\exp\left[-\frac{c^2+r^2}{2\sigma_d^2}\right]. \tag{37}$$

The filtered intensity is a weighted neighbourhood average, $\widetilde{I} = K_{\sigma_d} * I$. The interface offers Off, $\sigma_d = 0.5$, 0.7, and 1.0 pixels. Here σ_d describes the spatial width of the filter, whereas σ_n describes noise amplitude. They are different quantities with different units.

For independent additive intensity noise with variance σ_n^2 and a discrete normalized kernel K_{ij}, the filtered noise variance in an interior region is

$$\mathrm{Var}(\widetilde{n}) = \sigma_n^2 \sum_{i,j} K_{ij}^2. \tag{38}$$

Since the squared weights sum to less than one for a nontrivial averaging kernel, the variance is reduced. Adjacent filtered pixels become correlated, however, and the same convolution also smooths the signal. Larger σ_d therefore trades random-noise suppression for reduced localization detail and possible interpolation or deformation bias.

The implementation truncates the kernel at four standard deviations and uses a discrete radius $r_d = \lfloor 4\sigma_d + 0.5 \rfloor$: radii 2, 3 and 4 pixels for the three settings. Reflecting boundary handling is used for filtering, while support masks are eroded so that matching does not use footprints touching invalid material/background.

5.3 Local Wiener filtering

The implemented alternative is a spatially adaptive 5×5 local Wiener filter. If μ and v are the local mean and variance, and $\widehat{\sigma}_n^2$ is an estimated noise variance, the result is

$$\widetilde{I} = \mu + \max\left(0, 1 - \frac{\widehat{\sigma}_n^2}{v}\right)(I - \mu), \tag{39}$$

with a zero weight when the local variance is numerically zero. A low-variance neighbourhood is pulled towards its mean; a high-variance neighbourhood retains more original intensity variation. This is a local shrinkage rule, not an inverse optical-deblurring filter.

Noise estimation without clean-image access

Each capture is processed independently with a normalized second-difference high-pass kernel

$$H = \frac{1}{6}\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}. \tag{40}$$

Its squared entries sum to one. Over valid interior material pixels, the response $R = H * I$ supplies the robust estimate

$$\widehat{\sigma}_n = \frac{\text{median}\,|R - \text{median}(R)|}{0.6744897501960817}. \tag{41}$$

The simulated camera-noise input is not read by the filter. Fine speckle texture may itself contribute to the high-pass response, so the estimate is not exact. Equation (39) may also preserve noise where local texture variance is large or oversmooth weaker texture. These explain why adaptive filtering need not outperform a fixed Gaussian in DIC.

5.4 How filter quality is evaluated

Image appearance is not the final criterion. The relevant outcomes are displacement error, strain/stress error, accepted-point coverage and spatial resolution. A smoother image may look better while losing localization information. Conversely, a slightly blurred input may produce a more stable correlation fit in a noisy experiment.

The fresh comparison in Section 11 changes only the denoiser for each noise seed. It reports both each filter's accepted population and errors at the intersection of valid stress centres across all filters. This guards against an apparent improvement caused solely by rejecting difficult points. The study does not train or evaluate a learned denoising model.

6 Digital Image Correlation for Flat Specimens

6.1 Subset-based measurement

A subset is a square patch around a reference centre (c_0, r_0). Instead of following one pixel, DIC uses the intensity pattern of many neighbouring pixels to estimate local motion. An odd side length m gives symmetric offsets $-h, \ldots, h$, where $h = (m-1)/2$. Centres are spaced on a regular candidate grid. Grid spacing controls sampling density; subset size controls how much texture contributes to a match. These are independent settings.

The matcher receives only the images, their support masks and numerical settings. Physical material properties and analytical displacements are absent from its interface. The synthetic masks supply known geometric support, so the experiment remains conditional on that information rather than being fully blind to specimen geometry.

6.2 Integer-pixel initialization by ZNCC

At each eligible centre, an axis-aligned target window is searched within the configured radius. For reference intensities f_i and candidate target intensities g_i, zero-normalized cross-correlation is

$$C = \frac{\sum_i (f_i - \bar{f})(g_i - \bar{g})}{\sqrt{\sum_i (f_i - \bar{f})^2}\sqrt{\sum_i (g_i - \bar{g})^2}}. \tag{42}$$

Subtracting the mean removes an additive offset, and normalization removes a positive multiplicative brightness scale when clipping is absent. It does not remove random noise or spatially varying illumination. Nearly uniform reference subsets are rejected because their variance cannot provide reliable localization.

The code evaluates correlation using FFT-based correlations for patch sums, squared sums and the numerator. Crucially, candidate target windows whose *entire subset* lies outside valid processed support are excluded *before* selecting the best peak. If every candidate is unsupported, initialization fails explicitly. Otherwise the highest supported integer correlation supplies only the initial translation; the four displacement-gradient parameters initially equal zero.

This order matters near a hole. Selecting an invalid best peak and rejecting it afterwards would lose a possible valid second-best seed. Early support exclusion improves the available initialization, but does not prove that the selected supported peak is the correct material correspondence.

6.3 Six-parameter affine shape function

Let $(\Delta c_i, \Delta r_i)$ be offsets inside the reference subset. The deformed sampling locations are

$$c_i' = c_0 + \Delta c_i + p_0 + p_2 \Delta c_i + p_3 \Delta r_i, \tag{43}$$

$$r_i' = r_0 + \Delta r_i + p_1 + p_4 \Delta c_i + p_5 \Delta r_i. \tag{44}$$

Here (p_0, p_1) is centre translation and (p_2, p_3, p_4, p_5) represents local first derivatives of image displacement. The form accommodates translation, small rotation, extension and shear within the patch. It approximates higher-order spatial deformation and therefore has model error when a large patch crosses a strongly varying field.

Fractional target coordinates are sampled using `RectBivariateSpline`, a bicubic spline constructed from the final captured or filtered image values [6]. Subpixel displacement arises from optimizing continuous sample coordinates, not from increasing the original camera resolution.

6.4 Normalized nonlinear least-squares refinement

Define the normalized vectors

$$\widehat{\boldsymbol{f}} = \frac{\boldsymbol{f} - \bar{f}\mathbf{1}}{\|\boldsymbol{f} - \bar{f}\mathbf{1}\|}, \qquad \widehat{\boldsymbol{g}}(\boldsymbol{p}) = \frac{\boldsymbol{g}(\boldsymbol{p}) - \bar{g}(\boldsymbol{p})\mathbf{1}}{\|\boldsymbol{g}(\boldsymbol{p}) - \bar{g}(\boldsymbol{p})\mathbf{1}\|}. \tag{45}$$

The fitted parameters minimize the normalized sum of squared differences,

$$\min_{\boldsymbol{p}_\ell \le \boldsymbol{p} \le \boldsymbol{p}_u} \frac{1}{2}\|\boldsymbol{r}(\boldsymbol{p})\|^2, \qquad \boldsymbol{r}(\boldsymbol{p}) = \widehat{\boldsymbol{g}}(\boldsymbol{p}) - \widehat{\boldsymbol{f}}. \tag{46}$$

Since both normalized vectors have unit norm,

$$\|\boldsymbol{r}\|^2 = 2(1 - C), \qquad C = 1 - \tfrac{1}{2}\|\boldsymbol{r}\|^2. \tag{47}$$

Thus a low normalized squared difference and a high ZNCC describe the same fit criterion.

Locally, $\boldsymbol{r}(\boldsymbol{p} + \Delta\boldsymbol{p}) \approx \boldsymbol{r}(\boldsymbol{p}) + J_r \Delta\boldsymbol{p}$. The optimizer uses this local residual model with bounded steps to improve the parameters. The implementation calls SciPy's bounded nonlinear least-squares solver [7]; the flat matcher uses numerically approximated residual derivatives. It is not an implementation of inverse-compositional Gauss–Newton DIC. Translation bounds extend the search radius by 0.75 pixels and affine-gradient bounds are independently configured. Optimizer tolerances are 10^{-7} in the flat routine.

6.5 Acceptance and support after refinement

A numerically successful fit must also satisfy final correlation, parameter-bound and target-support checks. The final warped footprint is tested again because affine refinement can move or distort an initially supported square across a material boundary. Rejected estimates are retained for diagnostics where available, but are excluded from subsequent field recovery.

Status	Meaning in flat DIC
0	Accepted after all enabled checks
1	Reference subset crosses a processed mask or boundary
2	Insufficient reference intensity variation
3	Optimizer did not converge
4	Final ZNCC below the selected limit
5	No supported target seed or invalid final target support
6	Fitted parameter reached its configured bound
7	Independent reverse match failed
8	Reverse match returned to a different reference location

Table 2: Rejection diagnostics distinguish geometric exclusions from attempted matching failures.

6.6 Independent forward/backward verification

If a forward match maps $\boldsymbol{q}$ to $\widehat{\boldsymbol{q}}'$, the algorithm constructs a new source subset at $\widehat{\boldsymbol{q}}'$ in the deformed image and independently matches it back into the reference image. The reverse search is centred at that observed deformed location; it is not seeded with analytical motion or inverse forward-affine parameters. The reverse fit repeats texture, support, convergence, correlation and bounds checks.

The returned location $\widehat{\boldsymbol{q}}_{\text{return}}$ must satisfy

$$e_{\text{FB}} = \|\widehat{\boldsymbol{q}}_{\text{return}} - \boldsymbol{q}\|_2 \le 0.3\,\text{px} \tag{48}$$

by default. A failed reverse fit is status 7; an excessive finite return error is status 8. Batch settings may disable this check with a null tolerance for controlled comparisons. The images are not filtered again for the reverse fit.

What consistency does and does not mean

A reverse check catches some false correspondences and asymmetric failures. It is conservative near boundaries: an axis-aligned reverse source subset may be unsupported even if the forward warped footprint fits. It also adds optimization work. A periodic texture can return consistently to a wrong equivalent location, so a small cycle error is not a proof of uniqueness or an upper bound on physical displacement error.

6.7 Complete matching procedure

1. Prepare the processed image support and candidate-centre grid.
2. Reject reference subsets lacking full support or adequate texture.
3. Maximize integer ZNCC over fully supported target windows only.
4. Refine translation and gradients with the bounded affine image fit.
5. Apply forward convergence, correlation, footprint and bounds checks.
6. Independently reverse-match forward-accepted points and apply Equation (48).

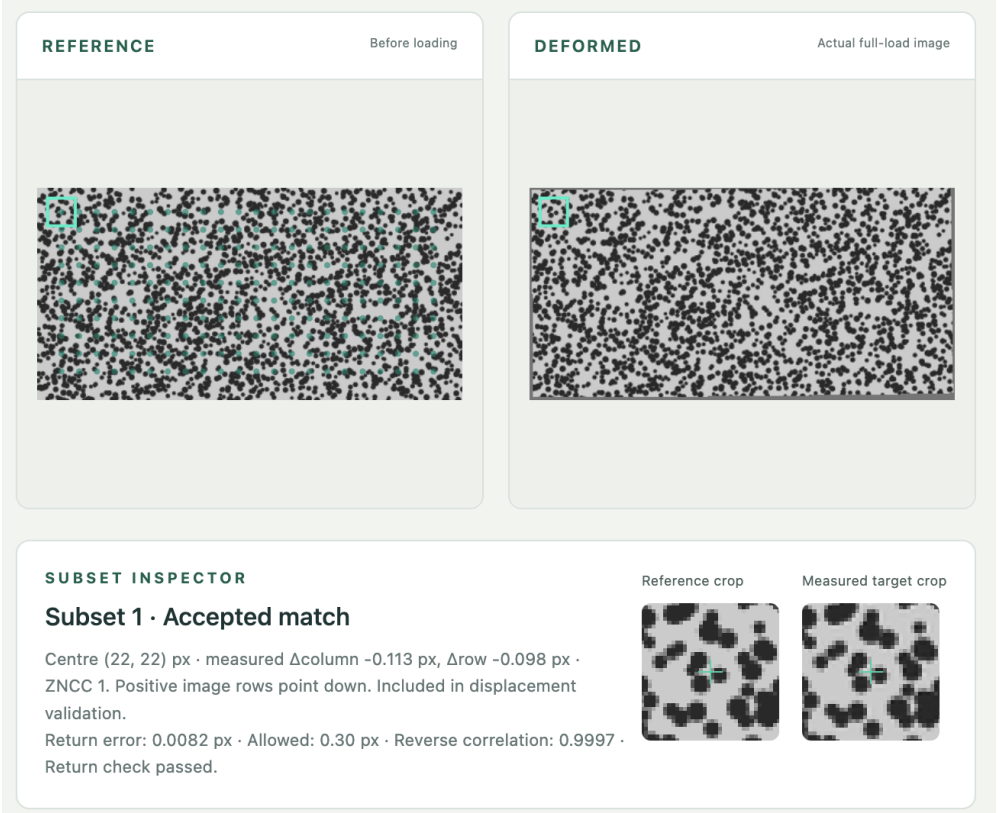


Figure: Cantilever reference and deformed speckle images with measurement locations and the subset inspector showing an accepted match and its forward/backward consistency check.

7 Recovering Strain and Inferring Stress

7.1 Why differentiating displacement is difficult

Displacement is the directly recovered kinematic field. Strain requires spatial derivatives, which amplify short-wavelength measurement noise. For independent displacement errors of variance σ_u^2, a two-point derivative over separation Δx has variance proportional to $2\sigma_u^2/(\Delta x)^2$. A very fine grid therefore does not by itself guarantee accurate strain. Local fitting uses a neighbourhood of accepted points to trade spatial resolution for noise suppression.

The reported strain is calculated from measured subset-centre displacements, not from analytical displacements and not directly from the affine-gradient parameters fitted inside individual subsets.

7.2 Weighted quadratic displacement fitting

For a centre (c_i, r_i) and fitting radius R_f in pixels, accepted neighbours within that radius are selected using a spatial search tree. Define dimensionless offsets

$$a_j = (c_j - c_i)/R_f, \qquad b_j = (r_j - r_i)/R_f. \tag{49}$$

The two image-displacement components are jointly fitted with

$$\begin{bmatrix} \widehat{d}_c(a,b) & \widehat{d}_r(a,b) \end{bmatrix} = \begin{bmatrix} 1 & a & b & a^2 & ab & b^2 \end{bmatrix} B. \tag{50}$$

Rows of the design matrix A contain these six basis functions. With $w_j = \exp[-(a_j^2 + b_j^2)]$, the code solves

$$\min_B \|W(AB - D)\|_F^2, \qquad W = \mathrm{diag}(w_j), \tag{51}$$

where D stores the measured image displacements. Since the rows are multiplied by w_j, the squared objective weights are w_j^2. The problem is solved with least squares, not by explicitly inverting $A^T W^2 A$.

At least six neighbours, full design-matrix rank and condition number no greater than 10^5 are required. An unavailable fit returns NaN. Six points alone are not sufficient if they fail to span the polynomial basis, for example when their geometry is nearly collinear.

7.3 Converting image derivatives into physical strain

The linear coefficients of Equation (50), divided by R_f, give $d_{c,c}$, $d_{c,r}$, $d_{r,c}$ and $d_{r,r}$. Using Equation (30), the physical small strains become

$$\varepsilon_{xx} = d_{c,c}, \qquad \varepsilon_{yy} = d_{r,r}, \qquad \gamma_{xy} = -\frac{s_x}{s_y} d_{c,r} - \frac{s_y}{s_x} d_{r,c}. \tag{52}$$

The two normal-strain conversions have cancelling scale factors. Shear retains their ratio as well as signs from the downward row coordinate. Assuming square pixels where $s_x \neq s_y$ would therefore give an incorrect shear conversion.

For the plate with a hole, neighbour links crossing the circular void are excluded. The intersection calculation is performed in the physical metric, with row offsets scaled by s_y/s_x. This avoids joining displacements through material that is absent.

7.4 Choosing the fitting radius

A larger radius averages more spatial information and often reduces random strain error. It can also smooth genuine field variation, distort a rapidly changing hole neighbourhood, or introduce bias when a quadratic is insufficient over the wider window. A smaller radius has higher spatial locality but can be poorly conditioned or too sensitive to displacement noise.

Changing only the strain-fit radius does not rerun or alter the DIC displacement fit. Consequently, displacement RMSE can remain identical while strain and stress RMSE change. Changing subset size is different: it affects the image-matching problem itself, eligible support and local deformation approximation. Neither setting has a universally monotonic effect on error.

7.5 Plane-stress constitutive conversion

For known E and ν, inversion of Equation (3) gives

$$\begin{bmatrix} \widehat{\sigma}_{xx} \\ \widehat{\sigma}_{yy} \\ \widehat{\tau}_{xy} \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix} \begin{bmatrix} \widehat{\varepsilon}_{xx} \\ \widehat{\varepsilon}_{yy} \\ \widehat{\gamma}_{xy} \end{bmatrix}. \tag{53}$$

Thus stress is *inferred* from measured strain through a constitutive assumption. Images alone cannot determine stress without such material information. The interface's material presets are illustrative elastic constants, not certified strength data. A strain error of 10^{-3} can correspond to a stress error of tens of MPa for a metal-like stiffness; low pixel error does not automatically imply low stress error.

7.6 Principal and equivalent stress

The two in-plane principal values are

$$\sigma_{1,2}^{\text{plane}} = \frac{\sigma_{xx}+\sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx}-\sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}. \tag{54}$$

The out-of-plane principal value is zero under plane stress; all three must be sorted if a three-dimensional principal ordering is needed. The plane-stress von Mises value is

$$\sigma_{\text{VM}} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\tau_{xy}^2}. \tag{55}$$

These scalar fields can be derived from recovered flat stresses. They are distinct from the report's combined-component stress-error statistic. No yield stress or failure criterion is supplied, so a red stress colour does not identify failure and the application does not compute a factor of safety.

8 Stereo DIC for Square-Beam Torsion

8.1 Why two-dimensional DIC is insufficient

Under torsion, a visible surface moves both within and out of the image plane. A single camera conflates perspective changes with in-plane motion and cannot uniquely reconstruct a general three-dimensional displacement. The project therefore uses two ideal calibrated cameras, each observing the reference and deformed surface: L_0, R_0, L_1 and R_1.

The mechanical model in Equation (28) supplies synthetic geometry only for image generation and later validation. The reconstruction module receives camera calibration, images, masks and its numerical settings. It does not receive torque, modulus, prescribed twist or analytical displacement. Only the common visible front face $z = a$ is measured; it is not a reconstruction of every external face or the solid interior.

8.2 Calibrated pinhole projection and image generation

For camera k, let its centre be $(C_{x,k}, 0, D)$, focal length be f pixels and principal point be $(c_{x,k}, c_y)$. A point (X, Y, Z) projects to

$$c_k = \frac{f(X - C_{x,k})}{D - Z} + c_{x,k}, \qquad r_k = c_y - \frac{fY}{D - Z}. \tag{56}$$

The cameras are rectified with parallel axes, no lens distortion, and an exactly known baseline $B = C_{x,R} - C_{x,L}$. Their principal points are adjusted to align the reference face in the two images. This exact simulated calibration eliminates a source of uncertainty that a physical stereo experiment would have to estimate.

Each output pixel is inversely projected onto the warped front surface. The code solves two projection equations for material coordinates (x, y) using a local two-by-two Newton update, with clipped increments and up to 24 iterations. Nonconverged, back-facing or out-of-bounds intersections are masked. The recovered material coordinates sample the reference texture with cubic interpolation. This geometric inverse projection is separate from both the flat fixed-point inverse warp and the image-correlation optimization.

8.3 Independent image correspondences

Each candidate starts at an integer reference location with full raw L_0 support. If denoising changes support, the original candidate population is retained so filtering failures remain visible. The local image matcher uses normalized affine fitting, the same supported-window initialization principle, a texture threshold of 0.015 and final ZNCC of at least 0.85.

Six image matches are evaluated:

1. $L_0 \to R_0$: reference stereo correspondence.
2. $L_0 \to L_1$: left temporal motion.
3. $L_0 \to R_1$: an independently measured right-deformed observation.
4. $L_1 \to R_1$: a second route to the right-deformed observation.
5. $L_1 \to L_0$: reverse temporal verification.
6. $R_0 \to R_1$: right temporal verification.

Previously measured image locations help centre some stereo/temporal searches, but no analytical torsion displacement supplies those guesses. Epipolar initialization restricts the row search for stereo pairs, while subsequent residual checks retain the measured row disagreement.

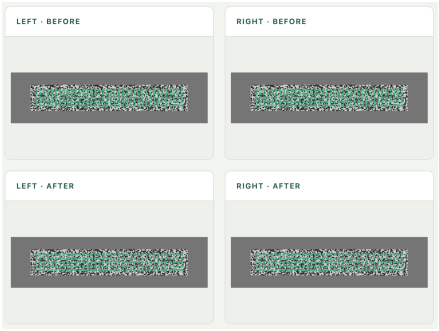


Figure: Four synthetic stereo camera captures for square-beam torsion: left and right reference images and left and right deformed images, with measurement locations.

8.4 Triangulation of reference and deformed surfaces

With observed left/right columns, define the principal-point-corrected disparity

$$\delta = (c_L - c_{x,L}) - (c_R - c_{x,R}). \tag{57}$$

The camera-depth distance is $q = fB/\delta$, so

$$Z = D - q, \tag{58}$$

$$X = \frac{1}{2}\left[C_{x,L} + \frac{(c_L - c_{x,L})q}{f} + C_{x,R} + \frac{(c_R - c_{x,R})q}{f}\right], \tag{59}$$

$$Y = \frac{\left[c_y - (r_L + r_R)/2\right]q}{f}. \tag{60}$$

Both the reference and deformed positions are triangulated. Their difference supplies the measured three-dimensional displacement.

Depth sensitivity follows directly from $q = fB/\delta$:

$$\frac{\mathrm{d}q}{\mathrm{d}\delta} = -\frac{fB}{\delta^2}. \tag{61}$$

A small disparity error can therefore cause a larger depth error. Low image-correlation residual alone does not establish accurate three-dimensional geometry.

8.5 Geometric and consistency checks

Triangulation must also be finite and have positive camera depth. The neighbour check predicts a candidate's image motion from surrounding accepted image measurements; it does not fit an imposed torsion shape. All failure reasons are retained. Rejection reduces coverage and can change the population used in RMSE.

8.6 Surface strain with rigid-rotation invariance

Let $\boldsymbol{r}(\eta_1, \eta_2)$ and $\boldsymbol{d}(\eta_1, \eta_2)$ denote the fitted reference and deformed 3D surfaces. A weighted quadratic fit is made to both surfaces using measured reference-image coordinates. At least nine neighbours, rank six and condition number no greater than 10^4 are required.

Check	Limit	Interpretation
Route/cycle disagreement	0.30 px	Reverse return and independently measured image routes agree
Epipolar row difference	0.25 px	Correspondences respect rectified camera geometry
Reprojection residual	0.20 px	Triangulated points project near measured image locations
Neighbour inconsistency	$\max(0.5\ \text{px}, 6s)$	Image motion agrees with a local affine neighbourhood prediction

Table 3: Existing stereo checks. Here s is a robust local residual scale; these limits are not calibrated physical uncertainty bounds.

Let $R = [\boldsymbol{r}_{,1}\ \boldsymbol{r}_{,2}]$ and $D_s = [\boldsymbol{d}_{,1}\ \boldsymbol{d}_{,2}]$ be their 3×2 tangent matrices, with the row axis reversed to correspond approximately to physical $+y$. Orthonormalize the reference tangents to obtain $Q = [\boldsymbol{e}_1\ \boldsymbol{e}_2]$, and set

$$A_r = Q^T R, \qquad F_s = D_s A_r^{-1}, \qquad E_s = \frac{1}{2}(F_s^T F_s - I_2). \tag{62}$$

The reported components are E_{11}, E_{22} and $2E_{12}$. The last is an engineering-style surface shear component. A poorly conditioned reference tangent transformation is rejected rather than inverted unreliably.

For a rigid rotation O, $D_s = OR$ and hence $F_s = OQ$. Since $O^T O = I$ and $Q^T Q = I_2$, Equation (62) gives zero strain. This is why the finite-rotation Green–Lagrange metric is appropriate for the twisting preview, whereas naively differentiating small-displacement components can report spurious strain under finite rotation.

8.7 Analytical validation and stress limits

The exact front-face tangent vectors obtained by differentiating Equation (28) provide the corresponding analytical surface metric. Comparing measured and analytical surface strains is therefore compatible with the actual camera geometry, including its finite section rotation.

Only surface kinematics have been reconstructed. The program does not infer all components of the interior stress tensor from this face. Cross-sectional shear and von Mises maps, twist curves and solid stress colours remain Saint-Venant analytical results. Their small-strain stress interpretation is maintained even though camera positions use finite rotation. Free warping is assumed; rigid end restraints and local end effects are excluded.

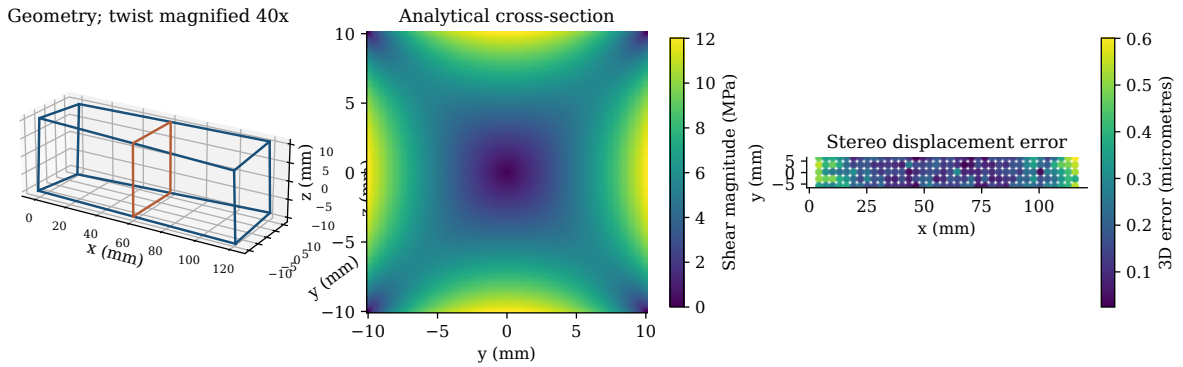


Figure 4: Square torsion: analytical opposite end rotations relative to the middle section, analytical cross-sectional shear, and measured visible-face displacement error. Bulk stress is analytical; the error map uses stereo reconstruction.

9 Implementation and Interactive Laboratory

9.1 Separation of responsibilities

The numerical modules are deliberately separated from presentation. This keeps image matching independent of the analytical deformation and makes exported results reproducible. Python performs mechanics, image processing, matching, recovery and validation. A local HTTP server accepts experiment settings and runs calculations in the background. Vanilla HTML, CSS and JavaScript present the controls and results; the interface does not require a JavaScript build system for normal use.

Module or path	Responsibility
`mechanics.py`	Compatible flat analytical displacement/stress fields
`speckle.py`	Texture generation, pixel calibration, inverse warp and camera model
`image_preprocessing.py`	Gaussian and local Wiener filtering with support margins
`dic.py`, `dic_support.py`	Supported integer initialization, affine matching and reverse checks
`strain.py`, `stress.py`	Local polynomial gradients and plane-stress inference
`validation.py`, `pipeline.py`	Errors, counts, orchestration and exports
`square_torsion.py`	Free-warping torsion mechanics and analytical solid geometry
`stereo_camera.py`	Calibrated synthetic stereo rendering and triangulation
`stereo_dic.py`	Image-only correspondence, checks and measured surface strain
`stereo_experiment.py`	Stereo orchestration, validation and output files
`lab_config.py`, `lab_server.py`	Input conversion, validation and interactive jobs
`web/`	Controls, deformation, image inspection, comparisons and readable results

Table 4: Core implementation modules. These describe implemented functions, not a proposed architecture.

9.2 User workflow and numerical limits

The user chooses one specimen, material, geometry and load, then selects camera and matching settings. Units displayed to the user are converted to SI before mechanics. The interface supports illustrative material presets and custom E, ν, camera noise/blur, brightness gain, speckle parameters, subset size, grid spacing, search radius, fitting radius and denoising.

Inputs are checked for finite values, geometry and solver feasibility. Flat interactive runs reject more than 2% predicted strain, excessive displacement gradients or motion beyond the selected search range. Square torsion limits the total twist to 0.15 rad and maximum shear strain to 2%, with constrained aspect ratios. These bounds limit the intended numerical/model regime; they do not check yielding or certify a real component.

9.3 Views and interpretation

- **Deformation:** analytical geometry and stress colours. Playback fraction scales the displayed load state, while visual magnification exaggerates geometry only.
- **Images and DIC:** original captures, filtered previews, subset locations and existing fit/reverse diagnostics.
- **Field comparison:** analytical, recovered and signed error maps on actual candidate locations, with shared truth/recovered colour scales.

- **Noise comparison:** optional paired noisy and no-added-noise flat runs evaluated at common valid centres.
- **Method and results:** matched/eligible counts, separate strain/stress counts, readable errors, rejection reasons and original technical JSON.
- **Stereo DIC:** four captures, visible-face 3D kinematics, surface-strain errors and existing stereo quality measures.

Changing settings marks previous results as stale until a new run completes. Measurement profiles select actual rows or columns; gaps preserve rejected data and hole crossings. Joining plotted samples does not create additional measurements. Noise comparison is a separate pair of full calculations, so it adds computational work.

9.4 Reproducible outputs

Each run records its configuration, summary, camera images, masks and measured data. Flat exports include displacement/status CSV, reverse-check CSV, mechanical fields, errors, dense ground truth and a visual report. Stereo exports preserve calibration and reconstructed fields. Invalid recovered entries remain NaN or unavailable rather than fabricated zeros. The selected raw and filtered inputs remain distinguishable.

10 Validation, Error Metrics and Robustness Protocol

10.1 Ground truth at matching locations

After matching, analytical fields are evaluated at each candidate's original reference position. A measured displacement at one centre is not compared with truth at a displaced location or with the nearest convenient point on a different mesh. The same location principle is applied to strain and stress.

For a component q, let $\mathscr{V}_q$ contain accepted points with finite measured and analytical values. The signed error, component RMSE and maximum absolute error are

$$e_i = \widehat{q}_i - q_i, \qquad \mathrm{RMSE}(q) = \sqrt{\frac{1}{N_q}\sum_{i\in\mathscr{V}_q} e_i^2}, \qquad e_{\max}(q) = \max_{i\in\mathscr{V}_q} |e_i|. \tag{63}$$

If $N_q = 0$, these errors are unavailable rather than zero. Each reported field retains its own valid sample count.

10.2 Vector and combined-component errors

For flat displacement errors measured in pixels,

$$\mathrm{RMSE}_{\boldsymbol{d}} = \sqrt{\frac{1}{N}\sum_{i\in\mathscr{V}} [(\widehat{d}_{c,i} - d_{c,i})^2 + (\widehat{d}_{r,i} - d_{r,i})^2]}. \tag{64}$$

The largest vector error is the maximum Euclidean norm over valid points. This differs from error in the scalar displacement magnitude, which would compare $\|\widehat{\boldsymbol{d}}\|$ with $\|\boldsymbol{d}\|$ and may hide directional error.

Flat strain and stress summaries analogously combine three stored components:

$$e_{\varepsilon,i}^2 = e_{\varepsilon_{xx},i}^2 + e_{\varepsilon_{yy},i}^2 + e_{\gamma_{xy},i}^2, \tag{65}$$

$$e_{\sigma,i}^2 = e_{\sigma_{xx},i}^2 + e_{\sigma_{yy},i}^2 + e_{\tau_{xy},i}^2. \tag{66}$$

These are numerical combined-component scores. They are not rotation-invariant tensor norms, because the stored engineering shear convention is not rescaled. In particular, combined stress error is *not* the error in von Mises stress. Conversion to microstrain multiplies strain values by 10^6; conversion to MPa divides pascal values by 10^6. Neither changes the underlying measured errors.

Stereo displacement uses the three-dimensional vector error in mm. Stereo surface-strain rows are individual component errors already expressed in microstrain. No unavailable combined maximum or recovered bulk-stress error is fabricated for that mode.

10.3 Coverage and the accepted population

Let N_c be all grid candidates, N_e eligible reference subsets, and N_a accepted displacement matches. The two useful fractions are

$$C_{\mathrm{all}} = N_a/N_c, \qquad C_{\mathrm{eligible}} = N_a/N_e, \qquad F_{\mathrm{eligible}} = 1 - C_{\mathrm{eligible}}. \tag{67}$$

C_{eligible} is undefined when $N_e = 0$. Reference-mask exclusions are $N_c - N_e$; matching failures are $N_e - N_a$. They answer different questions. Flat texture remains an eligible matching failure rather

than being relabelled a geometric exclusion. Stereo candidates initially require full raw reference support, and filtering or later checks can reject additional points.

Coverage states how many locations passed the procedure, not how accurate their values are. Rejection can decrease RMSE by removing difficult points without improving any surviving estimate. Conversely, a stricter consistency check can reject a correct-looking point and slightly increase the remaining RMSE. Both counts and errors must therefore be reported.

10.4 Hole-edge concentration versus sampled interior maximum

The report distinguishes three ratios: theoretical edge $K_t = 3$, analytical maximum σ_{xx}/S on valid recovered-stress centres, and recovered maximum $\widehat{\sigma}_{xx}/S$ on those *same* centres. A finite subset needs surrounding material and cannot be centred directly on the hole edge. The fitting window further averages nearby motion. An interior ratio below three is therefore not by itself evidence that the edge theory or the matcher is wrong.

10.5 Paired imaging and denoising comparisons

In a paired noise experiment, mechanics, texture, geometry, blur, gain and solver settings are unchanged; the baseline sets added Gaussian noise to zero. The baseline still includes interpolation, quantization and any selected blur. Errors use the intersection of finite accepted centres for each selected field. The exported counts remain separate.

The denoising comparison uses three independent camera-noise seeds with a fixed cantilever speckle realization and noise standard deviation 0.03. All five filters are compared at the intersection of valid stress centres within each repeat. This is a specific synthetic comparison, not a claim that one filter is universally best.

10.6 One-factor-at-a-time robustness study

The implemented default study uses the axial benchmark at 160×240 pixels, grid step 24 pixels and fitting radius 54 pixels. It repeats each factor level with camera seeds 253, 254 and 255. Four noise levels and three levels each of blur, brightness gain, subset size and speckle radius give $3(4+3+3+3+3) = 48$ trials. Other factors use background noise 0.01.

For each level the sample standard deviation is

$$s = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}(q_k - \bar{q})^2}, \qquad n = 3. \tag{68}$$

Error bars describe noise-seed variability under the fixed speckle realization, not confidence intervals or uncertainty over all patterns and material properties. Zero-noise repeats are deterministic. Changing subset size may change candidate and accepted centres; these default robustness RMSEs are not fixed-centre comparisons.

10.7 Independent software and mechanics checks

The repository's verification suite checks constitutive compatibility by displacement differentiation, beam resultants and free surfaces, hole-edge traction and concentration, known subpixel motion, mask support, noise independence, target initialization, reverse consistency, triangulation and rigid-rotation surface strain. Separate interface checks exercise control changes, exports and browser layout.

11 Numerical Results and Discussion

11.1 Baseline experiment protocol

All numerical tables and scientific plots in this section are synthetic calculations, not physical measurements.

The four flat baselines use $E = 70$ GPa and $\nu = 0.30$, speckle radius 2.2 px, density 0.4, texture seed 2039 and camera seed 253. Added noise and camera blur are zero; brightness gain is one; denoising is off. Subsets are 25×25 px, grid step is 16 px, search radius is 7 px and the strain-fit radius is 36 px. The flat reverse tolerance is explicitly 0.3 px.

Case	Observation size	Image $(H \times W)$	Load
Axial bar	120×80 mm	256×384	Remote stress $S = 280$ MPa
Cantilever	120×60 mm; $b = 6$ mm	224×448	Upward end force $P = 12000$ N
Plate with hole	80×80 mm; $a = 10$ mm	320×320	Remote stress $S = 140$ MPa
Pure bending	120×40 mm; $b = 4$ mm	160×384	End moment magnitude $M = 60$ N m

Table 5: flat-baseline configurations. Batch remote-stress inputs do not require an assigned bar/plate thickness. The metal-like constants and loads are benchmark inputs, not elastic-yield certification.

11.2 Flat displacement, strain and stress recovery

Specimen	Accepted / eligible	Displacement RMSE (px)	Strain RMSE ($\mu\varepsilon$)	Stress RMSE (MPa)
Axial bar	308/308	0.012747	333.0	26.310
Cantilever	312/312	0.015652	731.0	24.246
Plate with a hole	284/284	0.013404	267.0	15.285
Pure bending	176/176	0.014550	143.1	8.934

Table 6: Fresh baseline errors. Displacement is a two-component vector error; strain and stress are three-component numerical scores. All four cases have valid strain/stress at every accepted displacement centre in these runs.

Displacement RMSE is below 0.016 pixels for these four selected cases. Nevertheless, combined stress errors range from about 8.9 to 26.3 MPa. This is consistent with noisy numerical differentiation and multiplication by stiffness. The table does not establish the same errors for every load, image size or noise setting.

For the axial bar, true axial strain is $S/E = 0.004$, while recovered axial-strain component RMSE is approximately 0.000310. Comparing with this known nonzero scale is meaningful; dividing errors by zero transverse or shear ground truth would not be. Uniform stress does not mean the recovered pointwise estimates are uniform to machine precision.

In the cantilever, the engineering-shear component RMSE is approximately 0.000706, larger than either normal-strain component error. The local fit must differentiate a spatially varying displacement field, and the small-strain stress conversion carries this shear error into τ_{xy}. Figure 5 makes the recovered field and its residual visible separately.

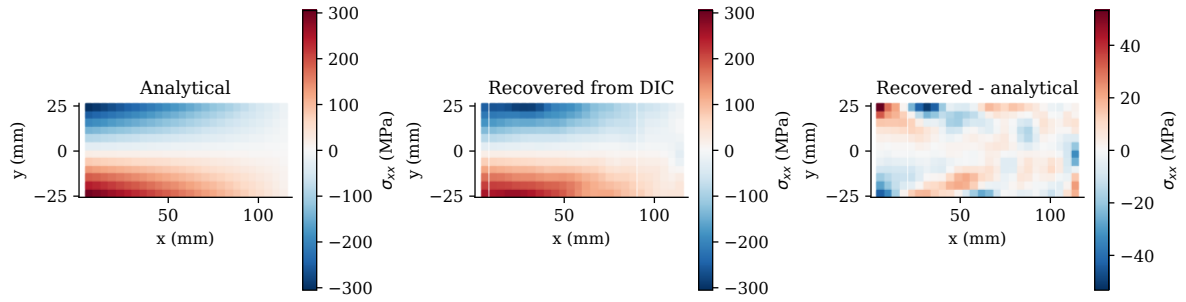


Figure 5: Cantilever axial stress at the same accepted subset centres: analytical, recovered and signed error. Truth and recovered panels share a scale; the error panel has its own symmetric scale.

Pure bending has quadratic displacement, which matches the order of the strain-recovery polynomial. It still has nonzero interpolation and image-matching errors. The maximum true displacement on sampled centres is only about 0.287 pixels, so a small absolute displacement error can matter substantially for inferred gradients. Increasing load to increase apparent motion must remain within the model and search limits.

11.3 The hole boundary and concentration interpretation

The plate has 324 grid candidates, 40 reference-mask exclusions and 284 accepted eligible matches. Its all-candidate acceptance is 87.65%, but its eligible matching coverage is 100%. Treating the 40 geometric exclusions as failed image matches would misstate the experiment.

For the fresh hole run, the theoretical edge factor is 3.0000, while the analytical sampled interior maximum is 1.5713 and the recovered sampled interior maximum is 1.4823. The last two use the same valid stress centres; they do not measure the edge peak. The visible gap around the hole in Figure 6 is a consequence of full subset support, not a filled region of zero stress.

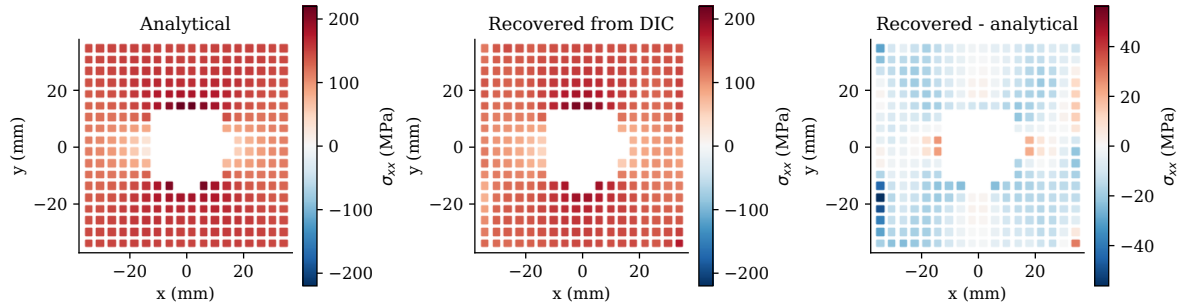


Figure 6: Plate-with-hole axial stress at accepted interior centres. The unmeasured hole and its support gap are retained. An interior sampled maximum must not be relabelled the analytical edge concentration factor.

11.4 Denoising: an experiment-dependent tradeoff

The cantilever denoising comparison adds Gaussian noise with standard deviation 0.03 and repeats camera seeds 253, 254 and 255. All other cantilever baseline settings remain fixed. Gaussian 0.5, 0.7, 1.0 px and local Wiener are compared with Off. Each repeat uses the common intersection of finite stress centres across all filters; the intersection sizes are retained in the trial data.

Gaussian 0.5 px gives the smallest common-centre displacement error in this selected study. Gaussian 1.0 px increases the error and reduces mean accepted points. Wiener and Off have similar common-centre stress scores, with Wiener slightly smaller here; three fixed-pattern repeats are too

Filter	Mean accepted displacement points	Common-centre displacement RMSE (px)	Common-centre stress RMSE (MPa)
Off	312.0	0.01558	40.04
Gaussian $\sigma_d = 0.5$	312.0	0.01413	40.20
Gaussian $\sigma_d = 0.7$	311.0	0.01576	46.11
Gaussian $\sigma_d = 1.0$	298.7	0.01881	58.89
Local Wiener	312.0	0.01596	39.95

Table 7: Three-repeat mean errors at identical valid centres within each repeat. Mean counts can be noninteger. Combined stress-component error is not von Mises error.

limited to establish a general ranking. Figure 7 includes trial points and sample deviations so the size of variation is visible.

These findings do not contradict the possibility that 1.0 px Gaussian denoising improves a noisier or differently loaded run. A filter modifies the texture and its interpolation behaviour as well as the noise. Matching error, acceptance and derivative bias depend on that full configuration. Selecting a filter solely because it makes the image smooth or because it won one earlier trial is therefore insufficient.

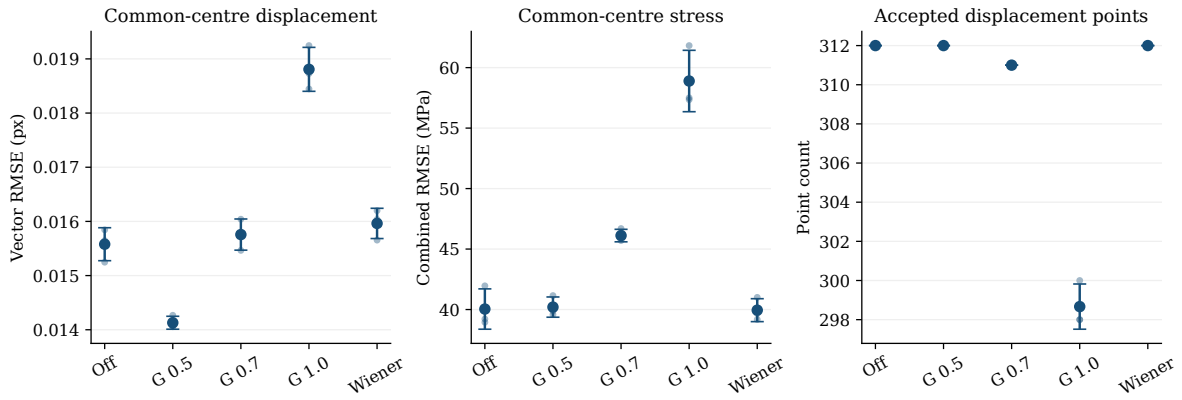


Figure 7: Controlled cantilever denoising comparison. G denotes Gaussian sigma in pixels. Left and middle errors use common finite stress centres; right counts show displacement acceptance. Error bars are sample standard deviations over three camera-noise seeds, with individual trials shown.

11.5 Imaging robustness across five factors

Figure 8 reports the 48 fresh default robustness trials. At each level, the curve is the mean accepted-point displacement RMSE over three seeds, with sample-standard-deviation bars.

The results should be interpreted as controlled sensitivity observations. Added noise alters the measured intensity pattern; blur changes its spatial bandwidth; gain is approximately normalized by ZNCC until clipping matters; subset size changes statistical support and spatial locality; speckle radius changes the available texture gradients. None of these factors is forced to produce a monotonic trend. Three zero-noise repeats coincide because only the unused noise seed changes.

11.6 Square torsion and visible-face stereo recovery

The square case uses length 120 mm, side 20 mm, torque 20 N m, $E = 70$ GPa and $\nu = 0.30$. It has 800-pixel-wide synthetic cameras, subsets 25 px, step 16 px, search radius 12 px and fitting radius 36 px. Noise, blur and denoising are off.

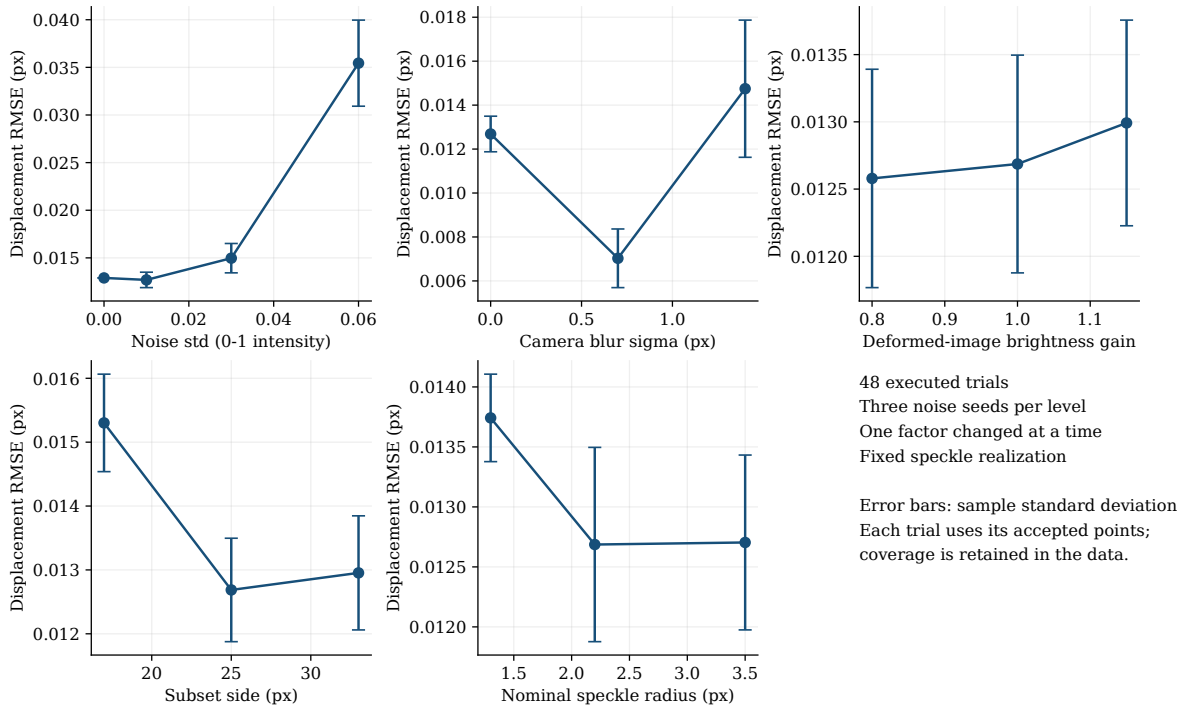


Figure 8: one-factor-at-a-time robustness study on the axial benchmark. Non-noise studies use background noise 0.01. Changing subset size can change eligible centres; these curves do not isolate a common fixed point population.

The fresh reconstruction accepts 190 of 190 displacement candidates and retains 186 surface-strain estimates. The 3D displacement vector RMSE is 0.00024151 mm, or about 0.242 micrometres, against a maximum analytical displacement of about 0.02142 mm on candidate centres. These very small values apply to ideal synthetic cameras with exact calibration, not to a physical camera system.

Measured component	Unit	Points	RMSE	Maximum error
Axial displacement	mm	190	6.80358e-05	0.000177391
Transverse displacement	mm	190	0.000209491	0.000599951
Depth displacement	mm	190	9.90549e-05	0.000305402
Surface axial strain	$\mu\varepsilon$	186	7.87155	44.8836
Surface transverse strain	$\mu\varepsilon$	186	20.4469	101.044
Surface shear strain	$\mu\varepsilon$	186	35.3347	102.503

Table 8: Fresh stereo component errors. Strain values compare the visible-face Green–Lagrange metrics. The reported surface shear component is $2E_{12}$, an engineering-style shear component. No DIC-derived bulk-stress is reported.

The reference/deformed triangulation and quadratic surface fits add stages beyond flat DIC. Their separate validation is important: high match coverage alone does not guarantee a well-conditioned surface gradient. Four centres therefore have displacement but no retained surface-strain fit. The cross-section stress in Figure 4 is analytical and must remain separate from this measured-face error table.



Figure: Comparison of analytical ground-truth and stereo-DIC recovered depth displacement on the visible reference face, together with the signed reconstruction error.

11.7 What the fresh evidence establishes

The selected baseline runs demonstrate subpixel recovery, consistent mechanical interpretation and explicit population accounting. The filter and robustness trials demonstrate that results depend on settings and that stronger filtering is not universally beneficial. They do not establish experimental accuracy, optimal tuning for every specimen, full-field recovery at free boundaries, or a recovered interior torsion stress tensor.

12 Limitations, Future Work and Conclusions

12.1 Model and measurement limitations

The flat mechanics are small-strain plane-stress benchmarks, not arbitrary three-dimensional structural solutions. The cantilever excludes local rigid-clamp effects; pure bending excludes moment-introduction effects; the hole case is an infinite-plate field viewed through a finite crop. Real material nonlinearity, yielding, buckling, plasticity and damage are absent.

Image generation uses prescribed texture transport, Gaussian optical blur/noise and 8-bit quantization. Real lighting, camera noise, lens distortion, surface reflectance, changing visibility and calibration uncertainty are more complex. Generation and matching use different spline implementations but related interpolation assumptions, so synthetic success is not independent physical validation.

The DIC solver is local and affine within a subset. There is no multiscale pyramid, global field regularization or demonstrated uniqueness test for repeated texture. Reverse consistency and stereo route checks reject some false matches but cannot solve unobservable periodic motion. Fitting windows introduce a spatial-resolution/noise tradeoff; complete subsets cannot measure directly at a hole edge. Reported stress requires the assumed material law and amplifies strain errors.

Stereo measures one visible front face with exact simulated calibration. Surface Green–Lagrange strain is recovered, while interior shear and stress colours remain analytical. A finite section rotation in camera geometry is not a geometrically nonlinear solid-stress solution.

12.2 Priorities for future development

Useful next steps are a broader controlled study of denoising and fitting-radius selection, calibrated ambiguity measures for repeated patterns, multiscale initialization for larger motion, and validation with real images and measured camera calibration. Learned denoising would require separate training/test patterns and evaluation by DIC error and coverage, not only image-quality scores. A general FEM extension would need its own meshing, boundary-condition treatment, equilibrium solver and verification; it is not obtained by changing the 3D renderer.

These are proposed extensions, not features claimed by the present report. Changes to measurement acceptance would require renewed numerical experiments so an improvement in surviving-point RMSE is not mistaken for an improvement over the full measurement population.

12.3 Conclusions

DIC Studio implements a transparent virtual link between compatible solid-mechanics fields and independent image-based deformation measurement. Four flat experiments recover in-plane displacement, strain and constitutively inferred stress. Square torsion extends the measurement to stereo reconstruction of visible-face 3D displacement and surface strain, with a middle-section rotation reference.

The fresh experiments demonstrate subpixel displacement recovery for the selected configurations, while showing appreciable strain and stress errors after differentiation. Explicit mask exclusions, acceptance diagnostics, valid-point counts and common-centre comparisons make those results interpretable. The denoising study shows that smoothing strength must be judged against the complete experiment rather than image appearance alone. The completed laboratory is therefore useful both for visualizing mechanics and for studying the limitations of the measurement process itself.

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A Notation and Units

Symbol	Meaning	Unit
x, y, z	Physical reference coordinates	m
$\boldsymbol{X}, \boldsymbol{x}$	Reference material and deformed spatial positions	m
u, v	Flat physical displacement components	m
c, r	Image column and row coordinates	px
s_x, s_y	Physical pixel sizes	m/px
d_c, d_r	Image displacement components	px
E, G	Young's modulus and shear modulus	Pa
ν	Poisson's ratio	dimensionless
P, M, T	Force, bending moment and torque	N, N m, N m
L, h, b	Beam length, height and thickness	m
a	Circular hole radius; for square torsion, square half-side	m
s	Square side length, with $s = 2a$ for square torsion	m
I, J	Bending second moment and torsion constant	m^4
S	Remote uniaxial stress	Pa
$\varepsilon_{xx}, \varepsilon_{yy}$	Small normal strains for flat specimens	dimensionless
γ_{xy}	Engineering shear strain $2\varepsilon_{xy}$	dimensionless
E_s	Surface Green–Lagrange strain tensor	dimensionless
σ_n	Camera Gaussian noise standard deviation	normalized intensity
σ_b, σ_d	Camera blur and Gaussian denoising widths	px
R_f	Strain fitting radius	px
α, θ	Twist rate and section rotation	rad/m, rad
$\mu\varepsilon$	Microstrain; one unit equals 10^{-6} strain	dimensionless